

Total Marks – 100
Attempt Questions 1-4

Question 1 (25 marks) Use a SEPARATE writing booklet. **Marks**

(a) Evaluate $\int_0^4 \frac{x \, dx}{\sqrt{16-x^2}}$. **2**

(b) Find $\int \sec^3 x \tan x \, dx$. **2**

(c) Evaluate $\int_0^2 \frac{dx}{x^2 - 4x + 8}$ **3**

(d) (i) Prove $2 \sin A \cos B = \sin(A-B) + \sin(A+B)$. **1**

(ii) Hence, evaluate $\int_0^{\frac{\pi}{12}} \sin 4x \cos 2x \, dx$. **2**

(e) (i) Find real numbers A, B, C such that

$$\frac{2x}{(1+x)(x^2+1)} = \frac{A}{1+x} + \frac{Bx+C}{x^2+1}.$$
 3

(ii) Hence find $\int \frac{2x}{(1+x)(x^2+1)} dx$. **2**

(f) (i) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. 2

(ii) Hence, evaluate $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$ 3

(g) (i) If $I_n = \int \frac{dx}{(x^2+1)^n}$ prove that

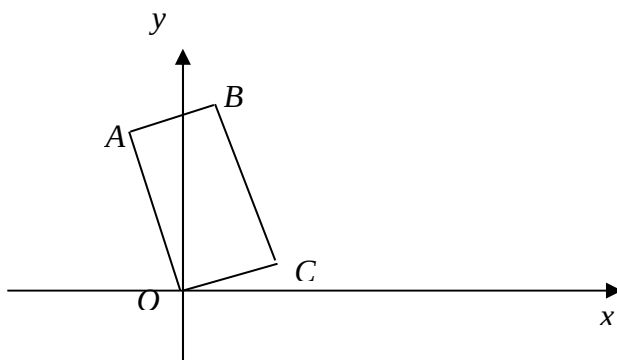
$$I_n = \frac{1}{2(n-1)} \left[\frac{x}{(x^2+1)^{n-1}} + (2n-3)I_{n-1} \right].$$
3

(ii) Hence evaluate $\int_0^1 \frac{dx}{(x^2+1)^2}$. 2

Question 2 (25 marks) Use a SEPARATE writing booklet.**Marks**

- (a) $z_1 = 1 + i$ and $z_2 = \sqrt{3} - i$.
- (i) Find $\frac{z_1}{z_2}$ in the form $a + ib$ where a and b are real. 1
- (ii) Write z_1 and z_2 in modulus – argument form. 2
- (iii) By equating equivalent expressions for $\frac{z_1}{z_2}$, write $\cos \frac{5\pi}{12}$ as a surd. 2
- (b) (i) Express in modulus argument form, $1 - \sqrt{3}i$. 1
- (ii) Hence evaluate $(1 - \sqrt{3}i)^5$ in the form $x + iy$. 3
- (c) Given $|z + i| \leq 2$ and $0 \leq \arg(z + 1) \leq \frac{\pi}{4}$. Sketch the region in an Argand diagram which contains the point P representing z . 3

(d)



The points $OABC$ are the vertices of a rectangle on the Argand diagram with $|OA| = 2|OC|$. If OC represents the complex number $p + iq$, write down the complex numbers represented by:

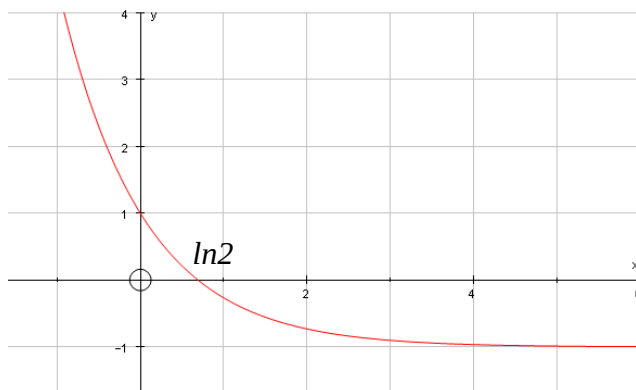
- (i) $\overrightarrow{OA}$ 1
- (ii) $\overrightarrow{OB}$ 1
- (iii) $\overrightarrow{BC}$ 1
- (iv) $\overrightarrow{AC}$ 1

- (e) Consider the five 5th roots of unity.
- (i) Solve $z^5 - 1 = 0$ over the complex field giving your answers in modulus-argument form. **3**
- (ii) Hence express $z^5 - 1$ as the product of real linear and quadratic factors. **3**
- (iii) Write down the complex roots of $z^4 + z^3 + z^2 + z + 1 = 0$ giving your answers in modulus-argument form. **1**
- (iv) Hence prove that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$. **2**

Question 3 (15 marks) Use a SEPARATE writing booklet.

Marks

(a)



The diagram shows the graph of $f(x) = 2e^{-x} - 1$. On separate diagrams sketch the following graphs, showing the intercepts on the axes and the equations of any asymptotes:

(i) $y = |f(x)|$. 2

(ii) $y = \{f(x)\}^2$. 2

(iii) $y = \frac{1}{f(x)}$. 2

(iv) $y = \ln \{f(x)\}$. 2

(v) $y = xf(x)$. 2

(b) Consider the curve $y^2 = x^4(4+x)$

(i) Sketch the curve. 2

(ii) Find the area of the loop of the curve from $x = -4$ to $x = 0$. 3

Question 4 (15 marks) Use a SEPARATE writing booklet. **Marks**

- (a) (i) Show that $(2x+1)$ is a factor of $P(x) = 2x^3 - 3x^2 + 8x + 5$ **1**
- (ii) Hence solve $P(x) = 0$ over the complex field. **3**
- (b) Consider the polynomial $P(x) = (x+2)^2 Q(x) + R(x)$
- (i) Explain why $R(x)$ is a linear polynomial **1**
- (ii) When $P(x)$ and $P'(x)$ are both divided by $(x+2)$, the remainder in each case is 6. Find $R(x)$. **3**
- (c) The roots of $x^4 + x^3 + 2x^2 + 3x + 1 = 0$ are $\alpha, \beta, \gamma, \delta$
- (i) Find the polynomial with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta}$. **2**
- (ii) Hence, or otherwise, show that **1**
- $$\left(\alpha + \frac{1}{\alpha}\right) + \left(\beta + \frac{1}{\beta}\right) + \left(\gamma + \frac{1}{\gamma}\right) + \left(\delta + \frac{1}{\delta}\right) = -4$$
- (d) (i) Explain why $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}$ has no double zeros. **2**
- (note: $3! = 3 \times 2 \times 1$)
- (ii) Find the relationship between a, b and c if $ax^4 + bx^2 + c$ has a double zero. **2**

End of paper

(a)

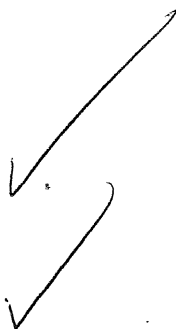
Q1

$$\int_0^4 \frac{x}{\sqrt{16-x^2}} dx$$

$$= - \left[\sqrt{16-x^2} \right]_0^4$$

$$= -(0 - 4)$$

$$= 4$$



(b)

$$\int \sec^3 x \tan x dx$$

$$= \int \sec^2 x \cdot \sec x \tan x dx$$

$$= \frac{\sec^3 x}{3} + C$$

Using Standard
Integral for
 $\int \sec x \tan x dx = \sec x$



(c)

$$\int_0^2 \frac{dx}{x^2 - 4x + 8}$$

$$= \int_0^2 \frac{dx}{(x-2)^2 + 4}$$

$$= \frac{1}{2} \left[\tan^{-1} \frac{(x-2)}{2} \right]_0^2$$

$$= \frac{1}{2} (0 - \tan^{-1}(-1))$$

$$= \frac{\pi}{8}$$



1(d) (i) Prove $2\sin A \cos B = \sin(A-B) + \sin(A+B)$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\therefore \sin(A-B) + \sin(A+B) = 2\sin A \cos B \quad \checkmark$$

$$(ii) \int_0^{\frac{\pi}{12}} \sin 4x \cos 2x \, dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{12}} [\sin(4x-2x) + \sin(4x+2x)] \, dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{12}} [\sin 2x + \sin 6x] \, dx$$

$$= -\frac{1}{2} \left[\frac{\cos 2x}{2} + \frac{\cos 6x}{6} \right]_0^{\frac{\pi}{12}}$$

$$= -\frac{1}{2} \left[\frac{\cos \frac{\pi}{6}}{2} + \frac{\cos \frac{\pi}{2}}{6} - \frac{1}{2} - \frac{1}{6} \right]$$

$$= -\frac{1}{2} \left[\frac{\sqrt{3}}{4} - \frac{2}{3} \right]$$

$$= \frac{8 - 3\sqrt{3}}{24}$$

$$(1+x)(x^2+1) \quad 1+x \quad x+1$$

$$2x \equiv A(x^2+1) + (1+x)(Bx+C)$$

$$\text{let } x=0$$

$$\therefore 0 = A + C$$

$$\text{let } x=1$$

$$\therefore 2 = -2 + 2(B+1)$$

$$\text{let } x=-1$$

$$-2 = 2A$$

$$\therefore B+1 = 2$$

$$B = 1 \checkmark$$

$$A = -1 \checkmark$$

$$\therefore C = 1 \checkmark$$

$$\therefore \frac{2x}{(1+x)(x^2+1)} \equiv \frac{-1}{1+x} + \frac{x+1}{x^2+1}$$

$$\begin{aligned} \text{(ii)} \quad \int \frac{2x}{(1+x)(x^2+1)} dx &= -\ln|1+x| + \int \frac{x}{x^2+1} dx + \int \frac{1}{1+x^2} dx \\ &= -\ln|1+x| + \frac{1}{2} \ln|x^2+1| + \tan^{-1}x \checkmark \end{aligned}$$

$$1 \text{ (8) (i)} \quad \int_0^a f(x) dx = \int_0^a f(a-x) dx \quad \text{let } u = a-x$$

$$\frac{du}{dx} = -1$$

$$\therefore \text{RHS} = - \int_a^0 f(u) \cdot \frac{du}{dx} dx \quad \checkmark$$

$$= - \int_a^0 f(u) du$$

$$= \int_0^a f(x) dx \quad \checkmark$$

$$= \text{LHS}$$

$$(ii) \quad \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$$

$$= \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x)}{1 + (\cos(\pi-x))^2} dx$$

$$= \int_0^{\pi} \frac{(\pi-x) \sin x}{1 + \cos^2 x} dx \quad \checkmark$$

$$\therefore 2 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx \quad \checkmark$$

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = -\frac{\pi}{2} \left[\tan^{-1}(\cos x) \right]_0^{\pi}$$

$$= -\frac{\pi}{2} \left[\left(-\frac{\pi}{4} \right) - \frac{\pi}{4} \right]$$

$$= \frac{\pi^2}{4} \quad \checkmark$$

$$I_n = \int \frac{dx}{(x^2+1)^n}$$

$$I_n = \int \frac{\frac{d(x)}{dx} \cdot dx}{(x^2+1)^n} \quad u=x, v=(x^2+1)^{-n}$$

$$= \frac{x}{(x^2+1)^n} + 2n \int \frac{x^2}{(x^2+1)^{n+1}} dx \quad \text{as } \frac{d}{dx} \left[\frac{x}{(x^2+1)^n} \right] = -\frac{2n}{(x^2+1)^{n+1}}$$

$$= \frac{x}{(x^2+1)^n} + 2n \int \frac{x^2+1}{(x^2+1)^{n+1}} - 2n \int \frac{1}{(x^2+1)^{n+1}} dx$$

$$= \frac{x}{(x^2+1)^n} + 2n I_n - 2n I_{n+1}$$

$$\therefore 2n I_{n+1} = \frac{x}{(x^2+1)^n} + (2n-1) I_n$$

$$\therefore I_{n+1} = \frac{1}{2n} \left[\frac{x}{(x^2+1)^n} \right] + \frac{2n-1}{2n} I_n$$

$$\text{let } n = N-1$$

$$\therefore I_N = \frac{1}{2(N-1)} \left[\frac{x}{(x^2+1)^{N-1}} + (2N-3) I_{N-1} \right]$$

$$\therefore I_n = \frac{1}{2(n-1)} \left[\frac{x}{(x^2+1)^{n-1}} + (2n-3) I_{n-1} \right]$$

$$(ii) \int_0^1 \frac{dx}{(x^2+1)^2} = I_2$$

$$\pi = \frac{1}{2} \left[\frac{x}{x^2+1} + \tan^{-1} x \right]_0^1 = \frac{\pi}{4}$$

$$(a) (i) \quad \frac{z_1}{z_2} = \frac{14i}{\sqrt{3}-i} \times \frac{\sqrt{3}+i}{\sqrt{3}+i}$$

$$= \frac{\sqrt{3}-1}{4} + \frac{(1+\sqrt{3})i}{4} \quad \checkmark$$

$$(ii) \quad z_1 = \sqrt{2} \operatorname{cis} \frac{\pi}{4} \quad \checkmark$$

$$z_2 = 2 \operatorname{cis} \left(-\frac{\pi}{6}\right) \quad \checkmark$$

$$(iii) \quad \therefore \frac{z_1}{z_2} = \frac{\sqrt{2} \operatorname{cis} \left(\frac{\pi}{4}\right)}{2 \operatorname{cis} \left(-\frac{\pi}{6}\right)}$$

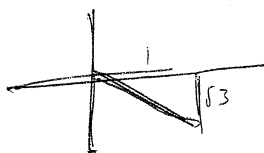
$$= \frac{1}{\sqrt{2}} \operatorname{cis} \frac{5\pi}{12} \quad \checkmark$$

$\therefore$ Equating real parts from (i) & (iii)

$$\frac{1}{\sqrt{2}} \cos \frac{5\pi}{12} = \frac{\sqrt{3}-1}{4}$$

$$\cos \frac{5\pi}{12} = \frac{\sqrt{6}-\sqrt{2}}{4} \quad \checkmark$$

$$(b) (i) \quad 1 - \sqrt{3}i = 2 \operatorname{cis} \left(-\frac{\pi}{3}\right) \quad \checkmark$$



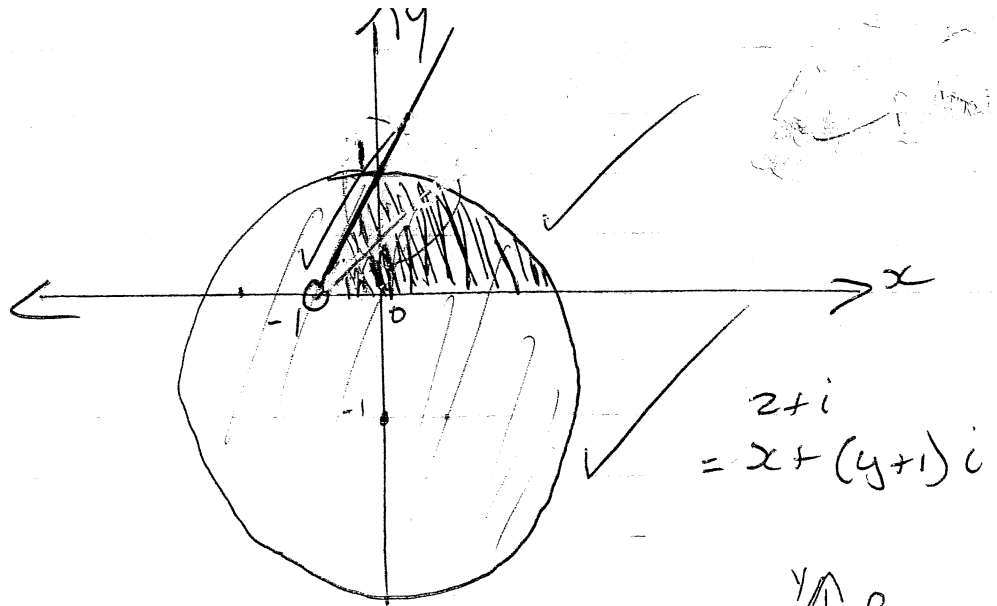
$$(ii) \quad (1 - \sqrt{3}i)^5 = 2^5 \operatorname{cis} \left(-\frac{5\pi}{3}\right) \quad \checkmark$$

$$= 2^5 \operatorname{cis} \frac{\pi}{3}$$

$$= 32 \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) \quad \checkmark$$

$$= 16 + 16\sqrt{3}i \quad \checkmark$$

Q2 (c)

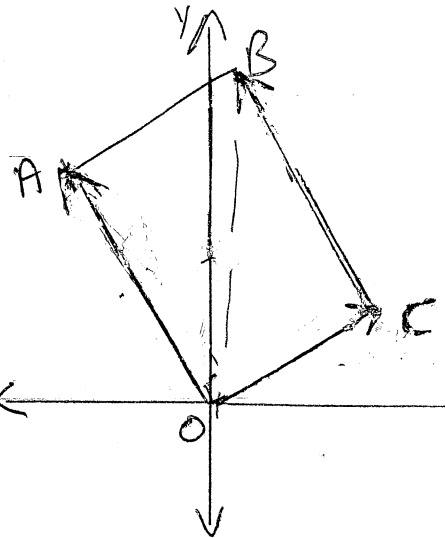


(d) Assuming $|OA| = 2|OC|$

(i) If OC is $p + iq$

$$\therefore \vec{OA} = 2 \times (p + iq) \times i$$

$$= -2q + 2ip$$



(ii) $\vec{OB} = \vec{OA} + \vec{OC}$

$$= (p - 2q) + (2p + q)i$$

(iii) $\vec{BC} = 2q - 2ip$

(iv) $\vec{AC}$

$$\vec{OC} + \vec{CA} = \vec{OA}$$

$$\vec{OC} - \vec{AC} = \vec{OA}$$

$$\vec{AC} = \vec{OC} - \vec{OA}$$

$$= p + iq - (-2q + 2ip)$$

$$= (p + 2q) + (q - 2p)i$$

2(e) (i) let

$$z = \cos \theta + i \sin \theta$$

$$z^5 = \cos 5\theta + i \sin 5\theta$$

$$\text{let } \cos 5\theta + i \sin 5\theta = 1$$

$$\therefore \cos 5\theta = 1$$

$$\therefore 5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$$

$$\theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$$

(iii)

$$\therefore 5 \text{ roots are } \cos 0, \boxed{\cos \frac{2\pi}{5}, \cos \frac{4\pi}{5}, \cos(-\frac{2\pi}{5}), \cos(-\frac{4\pi}{5})}$$

$$(ii) z^5 - 1 = (z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5)$$

$$= (z - 1)(z^2 - 2\cos \frac{2\pi}{5}z + 1)(z^2 - 2\cos \frac{4\pi}{5}z + 1)$$

$$\text{as Sum of conjugate root} = 2\cos \frac{2\pi}{5}, 2\cos \frac{4\pi}{5}$$

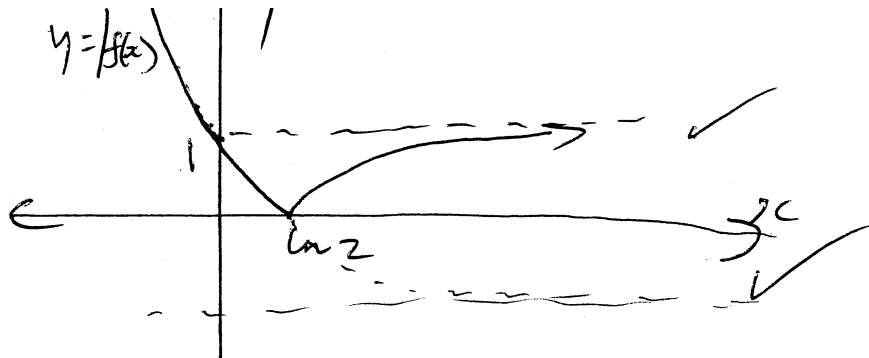
$$(iii) \text{ Sum of roots} = 0.$$

$$\therefore 1 + 2\cos \frac{2\pi}{5} + 2\cos \frac{4\pi}{5} = 0$$

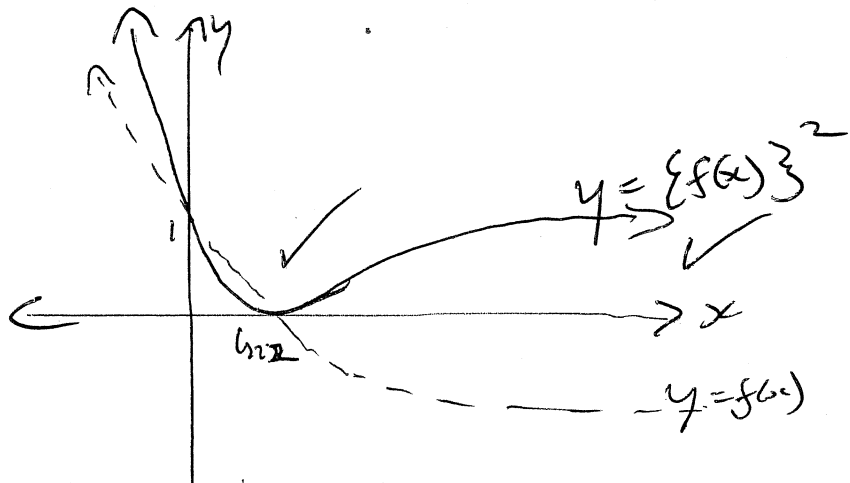
$$\therefore \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

(a)

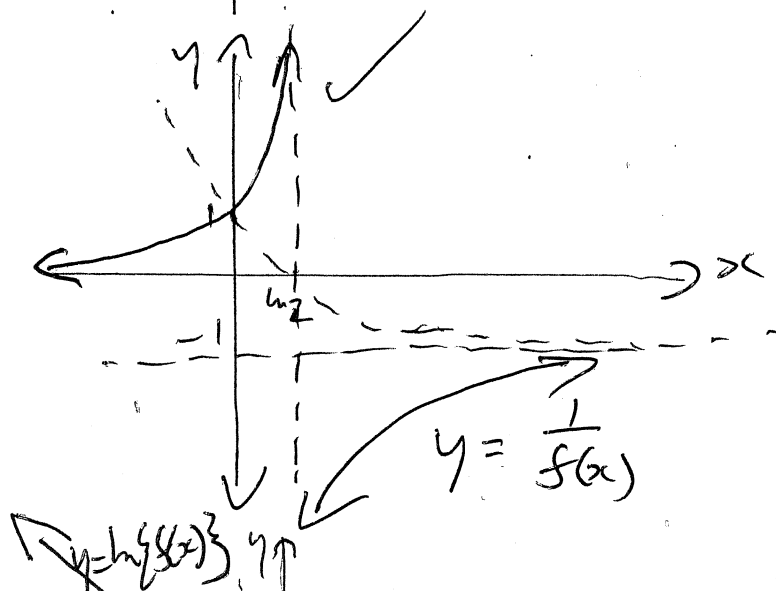
(i)



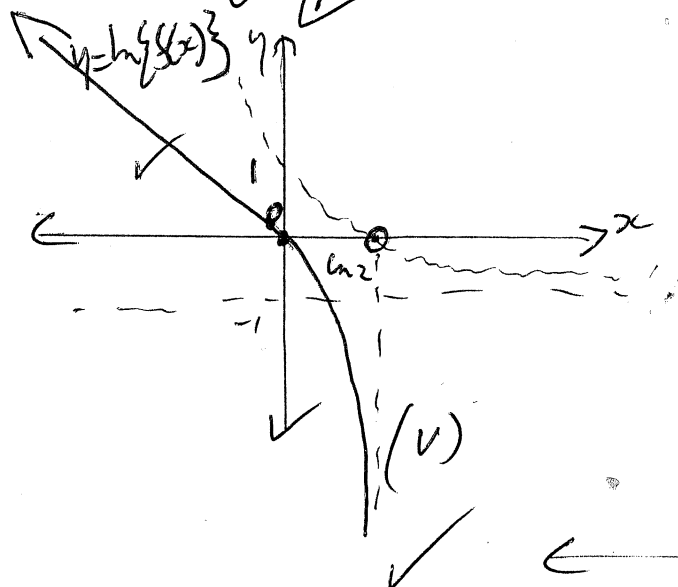
(ii)



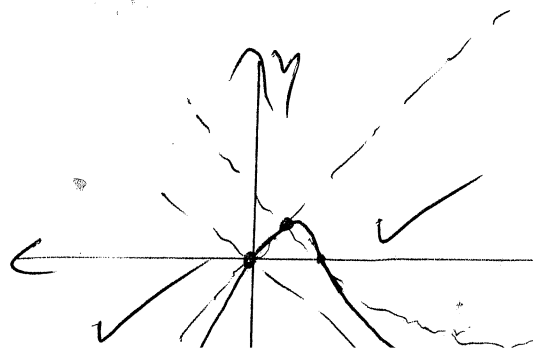
(iii)



(iv)

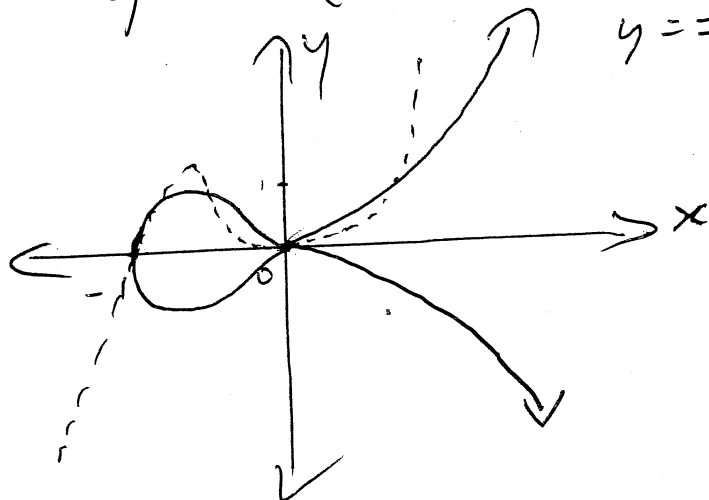


(v)



3(b)(i)

$$y^2 = x^4(4+x)$$



$$(ii) \text{ Area of loop} = 2 \int_{-4}^0 x^2 \sqrt{4+x} \, dx$$

$$\text{let } u = 4+x \quad \therefore \frac{du}{dx} = 1$$

$$\therefore \text{Area} = 2 \int_0^4 (u-4)^2 \sqrt{u} \, du$$

$$= 2 \int_0^4 (u^2 - 8u + 16) u^{\frac{1}{2}} \, du$$

$$= 2 \int_0^4 (u^{\frac{5}{2}} - 8u^{\frac{3}{2}} + 16u^{\frac{1}{2}}) \, du$$

$$= 2 \left[\frac{2}{7} u^{\frac{7}{2}} - \frac{16}{5} u^{\frac{5}{2}} + \frac{32}{3} u^{\frac{3}{2}} \right]_0^4$$

$$= \frac{4096}{105}$$

$$= 39 \frac{1}{105} u^2 \quad \#$$

$$Q4 (a) (i) P(-\frac{1}{2}) = -\frac{2}{8} - \frac{3}{4} - \frac{8}{2} + 5$$

$$= 0$$

$\therefore 2x+3$ is a factor

$$(ii) \begin{array}{r} 2x+1 \overline{) x^2 - 2x + 5} \\ \underline{2x^3 - 3x^2 + 8x + 5} \\ 2x^3 + x^2 \\ \underline{-4x^2 + 8x} \\ -4x^2 - 2x \\ \underline{10x + 5} \\ 10x + 5 \end{array}$$

✓

$$\begin{aligned} P(x) &= (2x+1)(x^2-2x+5) \\ &= (2x+1)[(x-1)^2+4] \\ &= (2x+1)[(x-1)^2-(2i)^2] \\ &= (2x+1)[x-1-2i][x-1+2i] \end{aligned}$$

$$\therefore P(x)=0$$

$$x = -\frac{1}{2}, \quad 1+2i, \quad 1-2i$$

$$4(5) \quad P(x) = (x+2)^2 Q(x) + R(x)$$

(i) $P(x)$ is divided by a quadratic $(x+2)^2$,
therefore the remainder must be in the form $ax+b$ ✓

$$(ii) \quad P'(x) = 2(x+2)Q(x) + Q'(x)(x+2)^2 + R'(x)$$

$$\therefore \text{as } R(x) = ax+b$$

$$R'(x) = a \quad \therefore a = 6$$

$$\text{Now as } P(2) = 6 \quad \therefore -2a+b = 6$$

$$\therefore b = 18$$

$$\therefore R(x) = 6x+18$$

$$(C) \quad x^4 + x^3 + 2x^2 + 3x + 1 = 0 \quad \dots (*)$$

$$(i) \quad \text{let } x = \frac{1}{x}$$

$$\therefore \left(\frac{1}{x}\right)^4 + 3\left(\frac{1}{x}\right)^3 + 2\left(\frac{1}{x}\right)^2 + 3\left(\frac{1}{x}\right) + 1 = 0$$

$$\therefore x^4 + 3x^3 + 2x^2 + x + 1 = 0 \quad \text{has roots } \frac{1}{x}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta}$$

$$(ii) \quad \text{Sum of roots } (*) = -1$$

$$\text{Sum of roots } \textcircled{*} = -3$$

$$\therefore \left(x + \frac{1}{x}\right) + \left(\beta + \frac{1}{\beta}\right) + \left(\gamma + \frac{1}{\gamma}\right) + \left(\delta + \frac{1}{\delta}\right) = -4$$

$$f'(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!}$$

Let $x = \alpha$ be the double root

$$\therefore 1 + \cancel{\alpha} + \frac{\alpha^2}{2} + \frac{\alpha^3}{8} = 1 + \cancel{\alpha} + \frac{\alpha^2}{2}$$

$$\therefore \alpha = 0$$

Now $f(0) \neq 0 \therefore$ No double root

(ii) Let $f(x) = ax^4 + bx^2 + c \quad \dots (1)$

$$\therefore f'(x) = 4ax^3 + 2bx \quad \dots (2)$$

From (2) $4ax^3 + 2bx = 0$

$$2x(2ax^2 + b) = 0$$

as $x \neq 0 \therefore x^2 = -\frac{b}{2a}$ ✓

$\therefore$ Sub into (1)

$$a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c = 0$$

$$\frac{ab^2}{4a} - \frac{b^2}{2a} + c = 0$$

$$b^2 - 2b^2 + 4ac = 0$$

$$4ac - b^2 = 0$$